

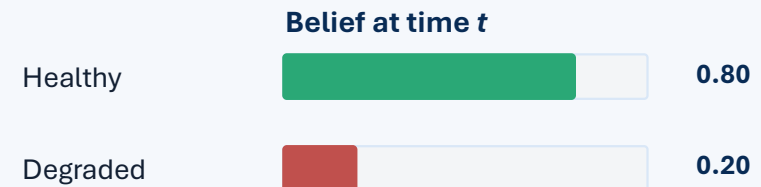
Artificial Intelligence

Lesson 22: Markov Models

State → Transition → Propagate → Repeat



State changes through time



A Markov model predicts how uncertainty about a system's state changes over time.



From static uncertainty to uncertainty through time

L18–21 gave us distributions and inference. L22 adds time.

Static belief

$$P(X)$$

What is likely now?

Belief with evidence

$$P(X | E)$$

What is likely given what we
observed?

Belief through time

$$P(X_{t+1} | X_t)$$

How does the state evolve?

New question: what if the uncertain state itself changes as time passes?

Today's mission

Given a current belief and transition model, predict future state probabilities.

- 1 Represent a first-order Markov process using a temporal graph, transition table, and transition matrix.
- 2 Interpret a transition matrix: current-state rows, next-state columns, and valid row probabilities.
- 3 Propagate a state distribution forward one or more time steps using the transition model.
- 4 Explain the Markov and stationarity assumptions and interpret long-run behavior.

Working rule: Read → Propagate → Interpret → Check

You have seen transition probabilities before

Markov models reuse the transition idea from MDPs, but without actions.

MDP transition model

$$P(s' \mid s, a)$$

current state + chosen action → distribution
over next states

Markov model transition

$$P(X_{t+1} \mid X_t)$$

current state → distribution over next states

Main difference: here we model state evolution, not action choice.

Temporal graph

Each node represents the same state variable at a different time.



$X_0, X_1, X_2, \dots$ are different time slices of the same system state.

- Example state: system health at each time step
- Possible values: Healthy or Degraded

$$X_t \in \{H, D\}$$

Start with a transition table

Before using a matrix, read the probabilities in ordinary table form.

Current state	Next state	Probability
Healthy	Healthy	.85
Healthy	Degraded	.15
Degraded	Healthy	.30
Degraded	Degraded	.70

$$X_t \in \{H, D\}$$

Read one row at a time

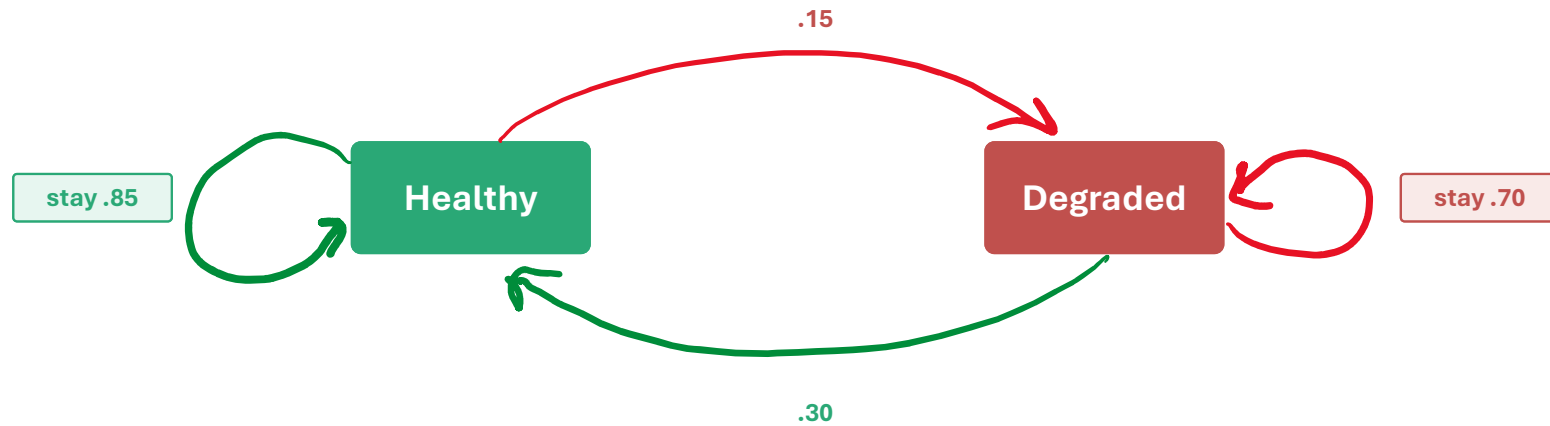
$$P(D_{t+1} | H_t) = .15$$

If the system is Healthy now, there is a 15% chance it is Degraded next.

A transition model describes a probability distribution over next states for each current state.

The same model as a state graph

Arrows show possible state changes and their probabilities.



Self-arrows represent persistence; cross-arrows represent switching states.

The same model as a transition matrix

A matrix compresses the transition table into a compact representation.

Transition matrix P		
	H_{t+1}	D_{t+1}
H_t	0.85	0.15
D_t	0.30	0.70

Orientation rule
Rows = current state
Columns = next state

$$P = \begin{bmatrix} 0.85 & 0.15 \\ 0.30 & 0.70 \end{bmatrix}$$

Do not read the matrix diagonally or by size. Read row $\rightarrow$ column.

Each row is a probability distribution

Given a current state, one of the next states must happen.

	H_{t+1}	D_{t+1}
H_t	0.85	0.15
D_t	0.30	0.70

Healthy row
 $.85 + .15 = 1$

Degraded row
 $.30 + .70 = 1$

A transition matrix is valid only if every current-state row sums to 1.

You Try: read the matrix

Read entries directly before doing any propagation.

	H_{t+1}	D_{t+1}
H_t	0.85	0.15
D_t	0.30	0.70

- $P(D_{t+1} | H_t) = ?$
- $P(H_{t+1} | D_t) = ?$
- If currently degraded, which next state is more likely?

If the current state is certain

A certain state can be written as a probability distribution.

Known current state

$$X_t = H$$

The system is Healthy now.

Distribution form

$$p_t = [1, 0]$$

100% Healthy, 0% Degraded

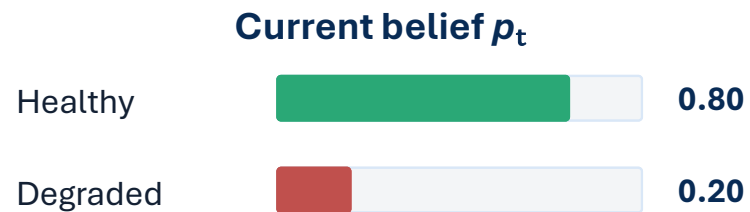
Next distribution is the Healthy row

	H_{t+1}	D_{t+1}
H_t	0.85	0.15
D_t	0.30	0.70

If $p_t = [1, 0]$, then $p_{t+1} = [.85, .15]$.

Usually the current state may be uncertain

The state at time t is often a belief distribution, not a known value.



Distribution notation

$$p_t = [.80, .20]$$

$$P(H_t) = .80 \text{ and } P(D_t) = .20$$

This is just another probability distribution—the same idea from L18.

Where can next-step degradation come from?

To predict D at the next time, consider every pathway into D.

Pathway 1

$$H_t \rightarrow D_{t+1}$$
$$P(H_t)P(D_{t+1}|H_t)$$

Pathway 2

$$D_t \rightarrow D_{t+1}$$
$$P(D_t)P(D_{t+1}|D_t)$$

$$P(D_{t+1}) = P(H_t)P(D_{t+1}|H_t) + P(D_t)P(D_{t+1}|D_t)$$

Multiply along each pathway, then sum the pathways.

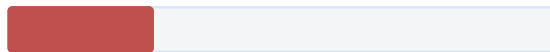
Compute the degraded probability

Use the current belief $p_t = [.80, .20]$ and the transition matrix.

$$\begin{aligned} P(D_{t+1}) &= P(H_t)P(D_{t+1}|H_t) + P(D_t)P(D_{t+1}|D_t) \\ &= (.80)(.15) + (.20)(.70) = .12 + .14 \end{aligned}$$

Next-step degraded probability

Degraded



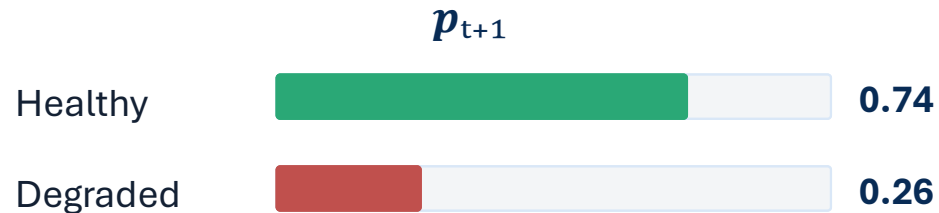
0.26

Interpretation: after one step, the model assigns 26% probability to being Degraded.

Compute the healthy probability

Use the same pathway idea for Healthy next.

$$\begin{aligned} & \text{Healthy next} \\ P(H_{t+1}) &= P(H_t)P(H_{t+1}|H_t) + P(D_t)P(H_{t+1}|D_t) \\ &= (.80)(.85) + (.20)(.30) = .68 + .06 \end{aligned}$$



Check: $.74 + .26 = 1$, so the result is a valid distribution.

Same probability pattern again

State propagation reuses the same probability operation from earlier lessons.

$$P(y) = \sum_x P(y | x) P(x)$$

L18

marginalization

L19

evidence term

L21

Bayesian inference

L22

state propagation

We are not learning a brand-new equation—we are reusing “weight pathways and sum.”

You Try: predict one step

Use probability pathways before using matrix notation.

	H_{t+1}	D_{t+1}
H_t	0.85	0.15
D_t	0.30	0.70

Start from

$$p_t = [.60, .40]$$

Find $P(H_{t+1})$ and $P(D_{t+1})$.

- Compute both next-state probabilities.
- Interpret the result in one sentence.
- Check that the results sum to 1.

Matrix notation packages the same calculation

Now that the pathway calculation is clear, we can compress it.

$$p_{t+1} = p_t P$$

$$\begin{bmatrix} 0.80 & 0.20 \end{bmatrix} \begin{bmatrix} 0.85 & 0.15 \\ 0.30 & 0.70 \end{bmatrix} \\ = \begin{bmatrix} 0.74 & 0.26 \end{bmatrix}$$

Matrix multiplication is a compact way to perform the pathway calculations we already understand.

Tiny matrix-multiplication primer

Each output entry uses one column of the transition matrix.

	H_{t+1}	D_{t+1}
H_t	0.85	0.15
D_t	0.30	0.70

Healthy output

$$(.80)(.85) + (.20)(.30) = .74$$

Degraded output

$$(.80)(.15) + (.20)(.70) = .26$$

$$p_{t+1} = p_t P = [0.80 \quad 0.20] \begin{bmatrix} 0.85 & 0.15 \\ 0.30 & 0.70 \end{bmatrix} = [0.74 \quad 0.26]$$

Rows tell where probability mass starts; columns tell where it lands.

You Try: apply the transition matrix

Use $p_{t+1} = p_t P$ to compute one complete step.

	H_{t+1}	D_{t+1}
H_t	0.85	0.15
D_t	0.30	0.70

Given

$p_t = [.30, .70]$
Compute p_{t+1} using matrix notation.

$$p_{t+1} = p_t P = [0.30 \quad 0.70] \begin{bmatrix} 0.85 & 0.15 \\ 0.30 & 0.70 \end{bmatrix} = [0.045 \quad 0.49]$$
$$= 0.535$$

Expected workflow: multiply into each output column, then check the distribution sums to 1.

More than one step: feed the answer back in

To predict farther ahead, use the transition model again.

After one step

$$p_{t+1} = [.74, .26]$$



Next step

$$p_{t+2} = [.74, .26]P$$

The output distribution becomes the next input distribution.

Compute the second step

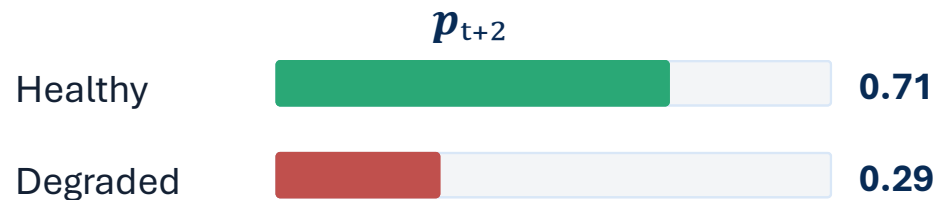
Apply the same transition matrix to $p_{t+1} = [.74, .26]$.

Healthy

$$(.74)(.85) + (.26)(.30) = .707$$

Degraded

$$(.74)(.15) + (.26)(.70) = .293$$



The model predicts a little more degradation after the second step.

Repeated propagation

The same operation can be repeated for many time steps.

$$p_{t+2} = (p_t P) P$$

$$p_{t+k} = p_t P^k$$

Interpretation

P^k is useful notation;
repeated application is the actual procedure to understand here.

You Try: two-step prediction

Start from a certain state and predict two steps ahead.

	H_{t+1}	D_{t+1}
H_t	0.85	0.15
D_t	0.30	0.70

Start from Degraded

$$p_0 = [0, 1]$$

Find $P(D_2)$.

- Step 1: compute p_1 .
- Step 2: feed p_1 back through the same transition matrix.
- Report the probability of Degraded at $t = 2$.

The Markov assumption

The current state should summarize the relevant effect of history.

$$P(X_{t+1} \mid X_t, X_{t-1}, \dots, X_0) = P(X_{t+1} \mid X_t)$$

Student-friendly version

Once the current state is known, older history provides no additional information needed for predicting the next state.

Markov is an assumption about state representation, not a claim that history is irrelevant.

Markov does not mean “history never mattered”

It means the current state carries the history that matters for prediction.

Wrong interpretation

“The past has no effect.”

This ignores how current state came to be.

Better interpretation

**“The relevant past is
encoded now.”**

The current state summarizes what we need
for the next prediction.

If older history would change the next-state probabilities, the state representation may be incomplete.

Quick Markov check

Does the proposed state representation contain enough information?

Model A

Tomorrow's condition depends only on today's condition.

Likely Markov under this representation.

Model B

Tomorrow's condition depends on today's condition and an unrecorded fault three days ago.

Not Markov unless that fault history is encoded now.

The repair is usually to improve the state representation—not to abandon the Markov idea.

Markov and stationary are different assumptions

Do not merge these two ideas.

Markov assumption

What history matters?

Given the current state, older history adds no predictive information.

Stationarity assumption

**Does the transition rule
change over time?**

The same transition matrix is reused at each time step.

A process can be Markov but not stationary if the transition probabilities change over time.

When stationarity may fail

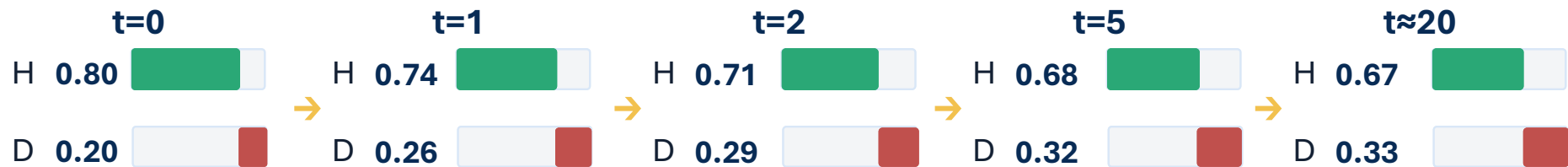
A single transition matrix may be too simple if the process itself changes.

- Equipment behaves differently under extreme heat.
- Network attack rate changes during an active incident.
- Weather transition probabilities differ by season.
- A component may degrade faster after known maintenance limits are exceeded.

Stationarity is useful—but it is an assumption that should be justified.

What happens if we keep applying the matrix?

Repeated transitions may cause the distribution to settle.



The influence of the initial distribution can fade for suitable chains.

Stationary distribution

A stationary distribution is unchanged by another transition.

$$\pi P = \pi$$

Translation

Once the distribution reaches π ,
another time step leaves the distribution unchanged.

We will interpret and verify stationary distributions, not solve them algebraically.

Verify — do not derive

Check a candidate stationary distribution by applying the matrix once.

Candidate

$$\pi = [2/3, 1/3]$$

	H_{t+1}	D_{t+1}
H_t	0.85	0.15
D_t	0.30	0.70

$$\begin{aligned}\pi P &= [2/3 \quad 1/3] \begin{bmatrix} .85 & .15 \\ .30 & .70 \end{bmatrix} \\ &= [2/3 \quad 1/3] = \pi\end{aligned}$$

If the distribution comes back unchanged, it is stationary.

Transfer: cyber host state

The same model applies to security state over time.

State variable
 $X_t \in \{\text{Secure}, \text{Compromised}\}$

Example transition matrix

	Secure _{t+1}	Comp _{t+1}
Secure _t	0.93	0.07
Comp _t	0.25	0.75

- Read one matrix entry.
- Compute one-step compromise probability from a given starting belief.
- State what the Markov assumption means in this cyber context.

Markov-model toolbox

Key pieces to carry into L23.

Piece	Meaning
p_0	initial state distribution
P	transition matrix
row	current state
column	next state
$p_{t+1} = p_t P$	propagate one step
Markov assumption	current state summarizes relevant history
stationarity	same transition model across time
stationary distribution	unchanged after another transition

If you can read P and propagate p , you can reason about temporal uncertainty.

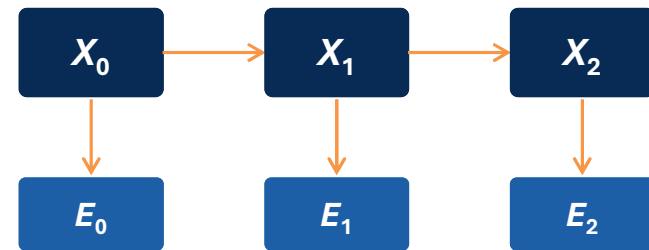
Bridge to Hidden Markov Models

What if the true state is hidden and we only receive noisy readings?

Markov model



Hidden Markov model



Next lesson

L23 adds observations:

Predict through the transition model, then update belief from noisy evidence.